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Electronic devices with sidewall displays

Abstract

Electronic devices may be provided that contain flexible displays that are bent to form displays on multiple surfaces of the devices. Bent flexible displays may be bent to form front side displays and edge displays. Edge displays may be separated from front side displays or from other edge displays using patterned housing members, printed or painted masks, or by selectively activating and inactivating display pixels associated with the flexible display. Edge displays may alternately function as virtual buttons, virtual switches, or informational displays that are supplemental to front side displays. Virtual buttons may include transparent button members, lenses, haptic feedback components, audio feedback components, or other components for providing feedback to a user when virtual buttons are activated.

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 18/304,158, filed Apr. 20, 2023, which is a continuation of U.S. patent application Ser. No. 17/973,658, filed Oct. 26, 2022, now U.S. Pat. No. 11,662,869, which is a continuation of U.S. patent application Ser. No. 17/588,072, filed Jan. 28, 2022, now U.S. Pat. No. 11,507,239, which is a continuation of U.S. patent application Ser. No. 17/164,611, filed Feb. 1, 2021, now U.S. Pat. No. 11,237,685, which is a continuation of U.S. patent application Ser. No. 16/720,257, filed Dec. 19, 2019, now U.S. Pat. No. 10,936,136, which is a continuation of U.S. patent application Ser. No. 16/421,892, filed May 24, 2019, now U.S. Pat. No. 10,521,034, which is a continuation of U.S. patent application Ser. No. 16/105,744, filed Aug. 20, 2018, now U.S. Pat. No. 10,318,029, which is a continuation of U.S. patent application Ser. No. 15/783,272, filed Oct. 13, 2017, now U.S. Pat. No. 10,055,039, which is a continuation of U.S. patent application Ser. No. 14/692,365, filed Apr. 21, 2015, now U.S. Pat.

No. 9,411,451, which is a continuation of U.S. patent application Ser. No. 14/602,199, filed Jan. 21, 2015, now U.S. Pat. No. 9,791,949, which is a continuation of U.S. patent application Ser. No. 14/273,315, filed May 8, 2014, now U.S. Pat. No. 8,976,141, which is a continuation of U.S. patent application Ser. No. 13/246,510, filed Sep. 27, 2011, now U.S. Pat. No. 8,723,824, all of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

(1) This relates generally to flexible displays, and more particularly, to electronic devices with flexible displays.

(2) Electronic devices such as portable computers and cellular telephones are often provided with rigid displays made from rigid display structures. For example, a liquid crystal display (LCD) may be formed from a stack of rigid display structures such as a thin-film transistor glass layer with display pixels for providing visual feedback to a user, a color filter glass layer for providing the display pixels with color, a touch screen panel for gathering touch input from a user, and a cover glass layer for protecting the display and internal components.

(3) Conventional devices may also have input-output components such as buttons, microphones, speakers, and other components that receive or transmit tactile input from a user mounted on edges of the device away from the display. Tactile input components are often formed from sliding or reciprocating button members and associated electrical components such as switches.

(4) Flexible display technologies are available that allow displays to be flexed. For example, flexible displays may be formed using flexible organic light-emitting diode (OLED) display technology.

(5) It would be desirable to be able to use flexible display technology to provide improved electronic devices such as electronic devices with input-output components.

SUMMARY

(6) Electronic devices may be provided with flexible displays. The flexible displays may include one or more flexible layers and may be mounted under a transparent display cover layer such as a layer of clear glass or plastic. For example, a flexible display may be mounted on the underside of a cover layer. Flexible displays may include a touch-sensitive layer that allows a user to provide touch input to an electronic device. Display pixels on a flexible display may be used to display visual information to the user.

(7) An electronic device may have a housing in which a flexible display is mounted. The housing and flexible display may be configured to form planar front and rear surfaces and sidewall surfaces for the device. A flexible display may be mounted so that at least a first portion of the flexible display is mounted on the front surface of the device and forms part of the front surface. The flexible display may have a bend that allows a second portion of the flexible display to cover some of the sidewall surfaces of the device.

(8) The flexible display may be used for displaying information and visual feedback to a user and for accepting input from a user. Active portions of the display configured for user input and output functions may be separated from inactive portions of the display using an opaque masking layer. The opaque masking layer may be formed on an inner surface of the cover layer.

(9) Openings may be formed in the opaque masking layer on the front and sidewall surfaces of the device. The front portion of the flexible display may be viewed through an opaque masking layer opening on the front of the device. Sidewall portions of the flexible display may be viewed through one or more sidewall openings in the opaque masking layer.

(10) Active portions (illuminated regions of pixels) on the sidewall edges of an electronic device may be used to create virtual user interface controls such as buttons. The buttons or other user input interface elements may be reconfigured during use of the electronic device. For example, the user input interface elements on the sidewall of an electronic device may be repurposed for supporting user input operations in different operating modes of the electronic device. Virtual buttons on the

edge of a device may be provided in place of tactile input/output components such as physical buttons and switches or may be formed as part of a dummy button structure or other mechanical feature.

(11) During operation of an electronic device, a virtual button may be, for example, a virtual volume button for controlling audio output volume and may be repurposed based on user input to become a virtual camera shutter button for taking a picture or may be reconfigured to serve as a controller for another device function. Images displayed on the flexible display may indicate to a user which function is currently being performed by the virtual button. Predetermined inputs to the touch-sensitive layer on the edge of the device (e.g., tapping, sliding, swiping, or other motions of an external object such as a finger across the edge of the device) may be used to change the operating mode of the device.

(12) Further features of the invention, its nature and various advantages will be more apparent from the accompanying drawings and the following detailed description of the preferred embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view of an illustrative electronic device with a flexible display having portions on multiple surfaces of a device in accordance with an embodiment of the present invention.

(2) FIG. **2** is a diagram of an illustrative set of display layers that may be used to form a flexible display in accordance with an embodiment of the present invention.

(3) FIG. **3** is a perspective view of a portion of an illustrative electronic device showing touch-sensitive edge displays on an edge of the device formed from a portion of a bent flexible display and a patterned housing member in accordance with an embodiment of the present invention.

(4) FIG. **4** is a perspective view of a portion of an illustrative electronic device showing touch-sensitive edge displays on an edge of the device formed from a portion of a bent flexible display and an opaque masking layer in accordance with an embodiment of the present invention.

(5) FIG. **5** is a perspective view of a portion of an illustrative electronic device showing touch-sensitive edge displays on an edge of the device formed from a portion of a bent flexible display in accordance with an embodiment of the present invention.

(6) FIG. **6** is a cross-sectional side view of a portion of an illustrative electronic device in the vicinity of a virtual button having a transparent button member in accordance with an embodiment of the present invention.

(7) FIG. **7** is a cross-sectional side view of a portion of an illustrative electronic device in the vicinity of a virtual button having an internal component in accordance with an embodiment of the present invention.

(8) FIG. **8** is a side view of an illustrative edge display portion of a flexible display showing how user input may be used to change the operating mode of an electronic device in accordance with an embodiment of the present invention.

(9) FIG. **9** is a side view of an illustrative edge display portion of a flexible display showing how portions of the edge display may be repurposed based on user input in accordance with an embodiment of the present invention.

(10) FIG. **10** is a cross-sectional side view of a portion of an illustrative electronic device in the vicinity of a virtual button having a transparent button member and a tactile feedback member in accordance with an embodiment of the present invention.

(11) FIG. **11** is a cross-sectional side view of a portion of an illustrative electronic device in the vicinity of a virtual button having an associated lens in accordance with an embodiment of the present invention.

(12) FIG. **12** is a cross-sectional side view of a portion of an illustrative electronic device in the vicinity of a virtual button having a transparent button member with a lens in accordance with an embodiment of the present invention.
(13) FIG. **13** is a diagram of illustrative control circuitry coupled to a touch-sensitive flexible display having a front surface display portion and a sidewall surface display portion with illuminated touch-sensitive regions in accordance with an embodiment of the present invention.
(14) FIG. **14** is a perspective view of an illustrative electronic device showing how virtual sidewall buttons may form a portion of a gaming controller when the device is operated in a landscape position in accordance with an embodiment of the present invention.
(15) FIG. **15** is a side view of a portion of an illustrative edge display portion of a flexible display showing how user the edge display portion may include virtual buttons for selecting a software application in accordance with an embodiment of the present invention.
(16) FIG. **16** is a side view of a portion of an illustrative edge display portion of a flexible display showing how user the edge display portion may include virtual buttons specific to a software application in accordance with an embodiment of the present invention.
(17) FIG. **17** is a side view of a portion of an illustrative edge display portion of a flexible display showing how user the edge display portion may include virtual buttons specific to a software application in accordance with an embodiment of the present invention.
(18) FIG. **18** is a side view of a portion of an illustrative edge display portion of a flexible display showing how user the edge display portion may include a scrollable list that is scrollable in multiple dimensions in accordance with an embodiment of the present invention.
(19) FIG. **19** is a side view of a portion of an illustrative edge display portion of a flexible display showing how user the edge display portion may display a different list based on a user touch-input to the edge display in accordance with an embodiment of the present invention.
(20) FIG. **20** is a side view of a portion of an illustrative edge display portion of a flexible display showing how user the edge display portion may display a different list entry based on a user touch-input to the edge display in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

(21) An electronic device may be provided that has a flexible display with bent edges. Bent edges of the flexible display may be visible along a sidewall or edge of the electronic device.
(22) Flexible displays may be formed from flexible layers such as a flexible display layer (e.g., a flexible organic light-emitting diode array), a flexible touch-sensitive layer (e.g., a sheet of polymer with an array of transparent capacitor electrodes for a capacitive touch sensor), a flexible substrate layer, etc. These flexible layers may, if desired, be covered by a flexible or rigid cover layer (sometimes referred to as a cover glass) or may be supported by a support structure (e.g., a rigid support structure on the underside of the flexible layers).
(23) Portions of the flexible display may be visible on multiple surfaces of an electronic device. For example, a planar portion of the display may be visible on a front or back surface of the device while an edge portion that has been bent along a sidewall of the electronic device may be visible on the edge of the device.
(24) Portions of the flexible display that are visible from the side of the device may be used to display information and virtual buttons for output and input of information to and from a user, respectively.
(25) An illustrative electronic device of the type that may be provided with a flexible display having bent edges visible along an edge of the device is shown in FIG. **1**. Electronic device **10** may be a portable electronic device or other suitable electronic device. For example, electronic device **10** may be a laptop computer, a tablet computer, a somewhat smaller device such as a wrist-watch device, pendant device, or other wearable or miniature device, a cellular telephone, a media player, etc.
(26) Device **10** may include a housing such as housing **12**. Housing **12**, which may sometimes be

referred to as a case, may be formed of plastic, glass, ceramics, fiber composites, metal (e.g., stainless steel, aluminum, etc.), other suitable materials, or a combination of these materials. In some situations, parts of housing **12** may be formed from dielectric or other low-conductivity material. In other situations, housing **12** or at least some of the structures that make up housing **12** may be formed from metal elements.

(27) Device **10** may have a flexible display such as flexible display **14**. Flexible display **14** may be formed from multiple layers of material. These layers may include a touch sensor layer such as a layer on which a pattern of indium tin oxide (ITO) electrodes or other suitable transparent electrodes have been deposited to form a capacitive touch sensor array. These layers may also include a layer that contains an array of display pixels. The touch sensor layer and the display layer may be formed using flexible sheets of polymer or other substrates having thicknesses of 10 microns to 0.5 mm or other suitable thicknesses (as an example).

(28) The display pixel array may be, for example, an organic light-emitting diode (OLED) array. Other types of flexible display pixel arrays may also be formed (e.g., electronic ink displays, etc.). The use of OLED technology to form flexible display **14** is sometimes described herein as an example. This is, however, merely illustrative. Flexible display **14** may be formed using any suitable flexible display technology. The use of flexible displays that are based on OLED technology is merely illustrative.

(29) Display **14** may have portions that are visible on a front side such as front surface **22** of device **10** and portions that are bent so that they are visible on edges such as sidewall surfaces **24** of device **10**. If desired, display **14** may be bent such that portions of display **14** are visible from a back side (e.g., a surface opposing front surface **22**) of device **10**.

(30) In addition to functional display layers (i.e., the OLED array and the optional touch sensor array), display **14** may include one or more structural layers. For example, display **14** may be covered with a flexible or rigid cover layer and/or may be mounted on a support structure (e.g., a rigid support). Layers of adhesive may be used in attaching flexible display layers to each other and may be used in mounting flexible display layers to rigid and flexible structural layers.

(31) In configurations for display **14** in which the cover layer for display **14** is flexible, input-output components that rely on the presence of flexible layers may be mounted at any suitable location under the display (e.g., along peripheral portions of the display, in a central portion of the display, etc.). In configurations for display **14** in which the flexible layers are covered by a rigid cover glass layer or other rigid cover layer, the rigid layer may be provided with one or more openings and electronic components may be mounted under the openings. For example, a rigid cover layer may have openings for components such as button **17** and speaker component **19** (e.g., for an ear speaker for a user). Device **10** may also have other openings (e.g., openings in display **14** and/or housing **12** for accommodating volume buttons, ringer buttons, sleep buttons, and other buttons, openings for an audio jack, data port connectors, removable media slots, etc.).

(32) Housing **12** may have openings such as openings **18** that allow bent portions of display **14** to be visible through openings **18**. In the example of FIG. **1**, housing **12** has three openings **18** for portions of display **14** that may be configured to be virtual buttons, virtual switches, scrolling displays, etc. This is merely illustrative. If desired, all of display **14** may be visible through housing **12** (e.g., using a transparent material to form housing **12**), housing **12** may have more than three openings, less than three openings, round openings, rectilinear openings, oval shaped or oddly shaped openings, etc. If desired, a transparent cover layer may extend over edges **24** of device **10** forming a continuous display around device **10**. Portions of a continuous display around device **10** may be configured to be virtual buttons, virtual switches, scrolling displays, etc. In configurations in which device **10** is provided with a continuous transparent cover layer, portions of display **14** may be separated from other portions of display **14** using a printed or painted mask on an internal surface of the cover layer or may be separated by selectively activating and inactivating display pixels to create virtual borders, virtual sections, or other visual delineations between portions of

display **14**.

(33) In some embodiments, portions of flexible display **14** such as peripheral regions **201** may be inactive and portions of display **14** such as rectangular central portion **20**A (bounded by dashed line **20**) may correspond to the active part of display **14**. In active display region **20**A, an array of image pixels may be used to present text and images to a user of device **10**. In active region **20**A, display **14** may include touch-sensitive components for input and interaction with a user of device **10**. If desired, regions such as regions **201** and **20**A in FIG. **1** may both be provided with display pixels (i.e., all or substantially all of the entire front planar surface of a device such as device **10** may be covered with display pixels). Edge portions of display **14** along edges **24** of device **10** may form a part of active regions **20**A. Edge portions of display **14** forming part of active region **20**A may contain portions of the array of image pixels for presenting to present text and images to a user of device **10** and touch-sensitive components for input and interaction with a user of device **10**.

(34) By folding the edges of flexible touch-sensitive display **14** (e.g., a flexible display layer and a flexible touch-sensitive layer), customizable illuminated touch-sensitive regions such as virtual buttons may be displayed along edges **24** of device **10**. Providing customizable virtual buttons may reduce system overhead costs and delays associated with creating and assembling individual physical buttons and switches.

(35) Customizable virtual buttons can be repurposed during normal operation of device **10**. Graphical and text displays on display **14** may indicate the current purpose and location of a virtual button to a user of device **10**. As an example, round virtual buttons indicating a “+” and “−” for raising and lowering audio output volume may be replaced by an image of a camera when a user changes from an audio mode of operation to an image capture mode of operation of device **10**. Virtual buttons may include buttons specific to a particular software application installed on device **10**. Virtual buttons may include locking and unlocking buttons. Locking and unlocking buttons may be operated using a swipe, pinch, or other touch action by a user of device **10**. Virtual buttons may include buttons specific to gaming software installed on device **10**. For example, virtual buttons may include buttons on edges **24** that may be operated by a user when holding device **10** in a landscape orientation during operation of device **10** in a gaming mode. Virtual buttons may be operated using touch, tap, swipe, pinch or other touch inputs to virtual buttons. Virtual buttons may include buttons commonly provided on a full sized computer keyboard (e.g., caps lock, shift, control, delete, page up/down, number lock, function specific buttons, escape, enter, etc.). Virtual buttons may include buttons commonly found on a calculator (e.g., multiply, add, divide, subtract, memory storage, clear, all clear, percent, square root, or other calculator buttons). Virtual buttons may include buttons for selecting specific software application available on device **10** (e.g., text messaging, calendar, calculator, media player, web browser, email client, cellular telephone, or other software applications). Virtual buttons may include images or icons that indicate the current function of the virtual button. Virtual buttons may include buttons commonly found on cellular telephone such as a menu button, a ringer on/off switch, a ringer on/off/vibrate switch, a lock/unlock button, a call button, an end-call button, or any other button associate with a cellular telephone.

(36) During normal operation, when virtual buttons are not needed, portions of display **14** that display virtual buttons along edges **24** of device **10** may be reassigned as an additional display for displaying text and image information to a user of device **10** or may be inactivated. Additional edge displays may be used to display scrollable lists such as artist lists, song lists, album lists, playlist lists, video lists, genre lists, webcast lists, audio book lists, or other scrollable lists. User touch input to edge displays may cause information to scroll vertically or horizontally across edge displays. As an example, in a media player mode of operation, a horizontal swipe may cause circuitry associated with device **10** to change an edge display from an artist list to a song list. A vertical swipe may cause circuitry associated with device **10** to scroll through a song list, an artist list, or other list. These examples are merely illustrative. Displays that are visible on edges of

device **10** may be used to display any information or to form any virtual button function.

(37) If desired, device **10** may include one or more sensors such as proximity sensor **124** for preventing erroneous inputs to virtual buttons such as virtual buttons **52** on a sidewall of device **10**. For example, in some modes of operation for device **10**, device **10** may be held in a portrait (e.g., vertical) orientation, while in other modes of operation for device **10**, device **10** may be held in a landscape orientation (e.g., a horizontal orientation). Holding device **10** in a portrait or landscape orientation may result in a user's hands covering different portions of device **10**. Sensors such as proximity sensor **124** may be used to determine whether a touch input to virtual buttons **52** is an intended touch by a finger or, for example, an unintended touch by the palm of a hand. This is merely illustrative. Other device components such as light sensors, motion sensors (accelerometers), capacitance sensors, etc. may be included and used to determine the orientation of device **10** and the intent of a touch input to virtual buttons **52**. Software running on device **10** may be configured to accept input from components such as proximity sensor **124** or other components to determine whether a touch-input to virtual buttons **52** is intended or unintended. Touch-inputs to virtual buttons **52** that are determined to be unintended may be ignored.

(38) An exploded perspective view of an illustrative display is shown in FIG. **2**. As shown in FIG. **2**, flexible display **14** may be formed by stacking multiple layers including flexible display layer **14**A, touch-sensitive layer **14**B, and a transparent display cover layer such as cover layer **14**C. Cover layer **14**C may form a planar front surface of device **10**. Cover layer **14**C may have a thickness of, for example, 0.1 mm to 3 mm, 0.1 to 1.5 mm, 0.1 to 2 mm, 1 to 2 mm, 0.7 to 2 mm, more than 0.1 mm or less than 2 mm. Flexible display **14** may also include other layers of material such as adhesive layers, optical films, or other suitable layers. Flexible display layer **14**A may include image pixels formed form light-emitting diodes (LEDs), organic LEDs (OLEDs), plasma cells, electronic ink elements, liquid crystal display (LCD) components, or other suitable image pixel structures compatible with flexible displays.

(39) Touch-sensitive layer **14**B may incorporate capacitive touch electrodes such as horizontal transparent electrodes **32** and vertical transparent electrodes **34**. Touch-sensitive layer **14**B may, in general, be configured to detect the location of one or more touches or near touches on touch-sensitive layer **14**B based on capacitive, resistive, optical, acoustic, inductive, or mechanical measurements, or any phenomena that can be measured with respect to the occurrences of the one or more touches or near touches in proximity to touch-sensitive layer **14**B.

(40) Software and/or hardware may be used to process the measurements of the detected touches to identify and track one or more gestures. A gesture may correspond to stationary or non-stationary, single or multiple, touches or near touches on touch-sensitive layer **14**B. A gesture may be performed by moving one or more fingers or other objects in a particular manner on touch-sensitive layer **14**B such as tapping, pressing, rocking, scrubbing, twisting, changing orientation, pressing with varying pressure and the like at essentially the same time, contiguously, or consecutively. A gesture may be characterized by, but is not limited to a pinching, sliding, swiping, rotating, flexing, dragging, or tapping motion between or with any other finger or fingers. A single gesture may be performed with one or more hands, by one or more users, or any combination thereof.

(41) Cover layer **14**C may be formed from or glass (sometimes referred to as display cover glass) or plastic and may be flexible or rigid. If desired, the interior surface of peripheral inactive portions **201** of cover layer **14**C may be provided with an opaque masking layer such as black ink, black plastic film, silver ink, silver plastic film or opaque masking layer of another color.

(42) Touch-sensitive flexible display section **14**AB may be formed from display pixel array layer **14**A and optional touch sensor layer **14**B.

(43) FIG. **3** is a perspective view of a portion of device **10** in the vicinity of openings **18** in housing **12**. As shown in FIG. **3**, cover layer **14**C may form a front side **22** of device **10**. Cover layer **14**C may be mounted to housing **12**. Housing **12** may have portions that form a sidewall **24** (also referred to herein as an edge or edge portion, housing sidewall, sidewall surface, etc.) of device **10**.

Edge portions **24** of device **10** may be substantially planar and may have portions that are substantially perpendicular to cover layer **14**C. Portions of housing **12** that form edges **24** of device **10** may be provided with openings such as openings **18**. Openings **18** may be partially or substantially filled by a transparent material such as transparent material **33**. Transparent material **33** may be formed form plastic, glass or any other suitable transparent material. Transparent material **33** may be flexible or rigid. Transparent material **33** may allow portions of touch-sensitive flexible display section **14**AB to be accessible through openings **18** in housing **12**. Touch-sensitive flexible display section **14**AB may be attached to housing **12** and cover layer **14**C using an adhesive layer such as adhesive layer **36**. Adhesive layer **36** may be formed from any suitable transparent adhesive. As shown in FIG. **3**, portions of display **14** may be visible on top side **22** of device **10** (e.g., a front side display) and portions of display **14** may be visible on edge **24** of device **10** (e.g., an edge display). Portions of display **14** visible on edge **24** of device **10** may be virtual user input-output components such as touch-sensitive edge display portions **52** (also sometimes referred to herein as touch-sensitive edge displays, virtual buttons, virtual interfaces, edge displays, edge interfaces, illuminated touch-sensitive display regions, or virtual switches). If desired, all or substantially all of edge **24** of device **10** may be used as an edge display. If desired control circuitry such as conductive traces **122**, may be formed in portions of display **14** that are hidden from view by housing **12**. Conductive traces **122** may provide control lines, drive lines, or other electrical connections for display pixels in display **14**.

(44) During manufacturing of device **10**, display pixels in display **14** that are positioned under openings **18** in housing **12** may be calibrated as button pixels (i.e., pixels that correspond to a virtual button such as virtual button **52**). During normal operation of device **10**, display pixels calibrated as button pixels may be configured to be illuminated and may be configured to display an image indicating the current function of the indicated pixels. User input (e.g., touch input using a finger) in the vicinity of button pixels of display **14** may activate the virtual button. A user of device **10** may change the function of button pixels by changing the operational mode of device **10**. Users may change the operational mode of device **10** using buttons such as button **17** of FIG. **1** or using virtual buttons on front side **22** or edge **24** of device **10**. User input that changes the operational mode of device **10** may be touch input (e.g., tapping, swiping, pinching, etc.) to touch-sensitive layer **14**B of flexible display **14**. Users may change the mode of operation of device **10** together with the display function of edge displays **52** or may change the mode display function of edge displays **52** without changing the mode of operation of device **10**. For example, during operation of device **10** in a cellular telephone or audio playback mode, edge displays **52** may function as virtual buttons for changing the volume of the audio output from device **10**. If desired, a user may change the display function of edge displays **52** to display the name of a caller or a song name, artist name, album name or other information related to a song, video or other media on device **10**. In some configurations, device **10** may be provided with multiple edge displays **52** as shown in FIG. **3**. In configurations in which device **10** includes multiple edge displays **52**, some edge displays **52** may be configured to operate as virtual buttons while other edge displays **52** are configured to operate as informational or graphical displays.

(45) As shown in FIG. **3**, flexible display **14** may be bent so that a portion such as portion **40** is parallel to front side **22** of device **10** and a portion such as portion **42** is parallel to edge **24** of device **10**. This is merely illustrative. Portions of flexible display **14** may, if desired, be parallel to any side of device **10** or may have a curved shape that conforms to non-planar portions of housing **12** or cover layer **14**C (e.g., convex or concave portions of device **10**).

(46) If desired, cover layer **14**C may be extended around a corner of device **10** from front side **22** to edge **24** of device **10** as shown in FIG. **4**. In the example of FIG. **4**, housing **12** and cover layer **14**C form an interface **48** on sidewall surface **24** of device **10**. Touch-sensitive flexible display section **14**AB may be attached to cover layer **14**C using an adhesive layer such as adhesive layer **36** interposed between touch-sensitive flexible display section **14**AB and cover layer **14**C. Portions of

an internal surface of cover layer **14**C may be patterned with a masking material to form a patterned opaque masking layer such as patterned opaque masking layer **46**. Patterned opaque masking layer **46** may be formed from any suitable masking material (e.g., black ink, silver ink, black or silver plastic film, etc.). Patterned opaque masking layer **46** may be painted, printed or otherwise deposited on inner surface **50** of cover layer **14**C so that active portions **20**A may be delineated from inactive portions **201** of touch-sensitive flexible display section **14**AB. As shown in FIG. **4**, one or more illuminated touch-sensitive regions (edge displays) **52** may be formed from active regions **20**A on edge **24** of device **10**. Patterned opaque masking layer **46** may be provided with openings such as openings **51** that define active display regions **20**A on front surface **22** and sidewall surface **24** of device **10**. Portions of display **14** may be visible through openings **51**. Patterned opaque masking layer **46** may have portions interposed between portions of display cover layer **14**C on front side **22** of device **10** and flexible display **14**. Patterned opaque masking layer **46** may have portions interposed between portions of display cover layer **14**C on sidewall surface **24** of device **10** and flexible display **14**. Patterned opaque masking layer **46** may have openings **51** under display cover layer **14**C on both front side **22** and sidewall surface **24** of device **10**. Touch-sensitive flexible display layers **14**AB may be visible through openings **51**. Display pixels in touch-sensitive flexible display section **14**AB in active regions **20**A on edge **24** may be pre-calibrated during manufacturing of device **10** as button pixels associated with virtual interfaces **52**. Virtual interfaces **52** may be virtual buttons (e.g., for raising or lowering audio volume, for activating an electronic or mechanical camera shutter, for changing operational modes, etc.), may be virtual switches, or may be supplemental displays for displaying text, image, video or other information for users of device **10**. Active region **20**A on front side **22** of device **10** may form a front surface display portion of display **14** and illuminated touch-sensitive regions **52** on sidewall **24** of device **10** may be formed from a sidewall surface portion of display **14**. The front surface display portion may be visually separated from illuminated touch-sensitive regions **52** using patterned opaque masking layer **46**. If desired control circuitry such as conductive traces **122**, may be formed in portions of display **14** that are hidden from view by patterned opaque masking layer **46**. Conductive traces **122** may provide control lines, drive lines, or other electrical connections for display pixels in display **14**.

(47) If desired, active portions **20**A and inactive portions **201** of display **14** may be defined using active portions **54** and inactive portions **56** of touch-sensitive flexible display section **14**AB as shown in FIG. **5**. In inactive portions **56** of touch-sensitive flexible display section **14**AB, display pixels in display layer **14**A and touch-sensitive elements in touch-sensitive layer **14**B may be temporarily or permanently disabled. Inactive portions **56** and active portions **54** of touch-sensitive flexible display section **14**AB may be configured to create edge displays **52** on edge **24** of device **10**. In configurations in which edge displays **52** are created using active portions **54** and inactive portions **56** of touch-sensitive flexible display section **14**AB, virtual user interfaces **52** may be re-positioned, resized, or otherwise reallocated by changing the distribution of inactive and active display pixels and touch-sensitive elements. As an example, one or more virtual buttons **52** may be moved and resized to position **60** along edge **24** of device **10**. Alternatively, an additional virtual button **52** may be added at position **60** on sidewall surface **24** of device **10**.

(48) Touch-sensitive flexible display section **14**AB may be attached to cover layer **14**C using any suitable transparent adhesive **36**. Providing device **10** with a flexible display such as flexible display **14** that conforms to inner surface **50** of cover layer **14**C may allow substantially all of front side **22** and edge **24** of device **10** to be part of active display area **20**A and to be used for display and user interface purposes. Active regions **20**A on edge **24** of device **10** may be used as a display that is supplemental to active portions **20**A on front side **22** of device **10**. Supplemental displays on edge **24** of device **10** may be virtual user interface components, scrolling displays, may display information about media (e.g., songs or movies) currently playing or currently stored on device **10**, may display information about current or recent cellular telephone calls, text messages, email

updates, webpage updates, etc.

(49) Portions of display **14** may interchangeably be allocated to active regions **20**A and inactive regions **201**. Portions of display **14** that form virtual buttons during one mode of operating device **10** may be repurposed to form a portion of an informational display during another mode of operation of device **10**. Virtual buttons that are created in a portion of display **14** that is repurposed may be reallocated to another portion of display **14**. If desired control circuitry such as conductive traces **122**, may be formed in inactive region **201**. Conductive traces **122** may provide control lines, drive lines, or other electrical connections for display pixels in display **14**.

(50) As shown in FIG. **6**, virtual buttons **52** on edge **24** of device **10** may have an associated transparent button member. Virtual buttons **52** formed from active portions of touch-sensitive flexible display section **14**AB may be formed in openings such as openings **18** in a structural member such as structural member **68**. Structural member **68** may be formed from a portion of housing **12**, may be formed from a portion of cover layer **14**C or may be formed from another structural member of device **10**. A button member such as button member **62** may be mounted in opening **18**. Button member **62** may be a dummy button member that serves as a tactile indicator for a user of device **10** indicating the location of virtual button **52**. Button member **62** may be formed from a transparent material such as plastic, glass, or other transparent material. Light such as light **64** generated by display pixels in display layer **14**A may pass through button member **62** so that virtual button **52** may be visible through button member **62**. Providing virtual button **52** with a transparent button member such as button member **62** may provide a user with a tactile button indicator while allowing virtual button **52** to be repurposed for different operating modes of device **10**.

(51) As shown in FIG. **7**, virtual buttons **52** on edge **24** of device **10** may have an associated feedback element for providing tactile and/or audio feedback to a user of device **10** during activation of virtual buttons such as virtual buttons **52**. As shown in FIG. **7**, feedback components such as feedback component **70** may be mounted to flexible display **14** in the at the location of one or more of virtual interfaces **52**. Feedback component **70** may generate a haptic, audio or other feedback response when virtual button **52** is activated. Virtual button **52** may be activated by a response to a touch or near touch in the vicinity of an active portion of touch-sensitive layer **14**B of display **14** associated with virtual button **52**. Feedback component **70** may be an actuator such as a motor, solenoid, vibrator, or piezoelectric actuator, an audio component such as a speaker, or other component. In configurations in which cover layer **14**C is rigid, feedback component **70** may be an audio feedback component such as a speaker that produces a sound when virtual button **52** is activated. In configurations in which cover layer **14**C is flexible, component **70** may contain an actuator such as a piezoelectric actuator **70**. Piezoelectric actuators may vary in shape (e.g., thickness) in response to applied control voltages and may produce an output voltage when compressed (i.e., the piezoelectric element in component **70** may serve as a force sensor in addition to serving as a controllable actuator). A user of device **10** may exert force on flexible display **14** in direction **72**. If desired, flexible display **14** may be deformed to exert a mechanical pressure on component **70**, inducing a voltage which may be transmitted to device **10**. Conversely, component **70** may be used to provide tactile feedback to a user of device **10**. A voltage difference applied to the surfaces of component **70** may induce an expansion of a piezoelectric actuator. Component **70** may then deform flexible display **14** in a direction outward of device **10** providing tactile feedback to a user of device **10**.

(52) FIG. **8** shows a how portions of an edge display such as edge display **52** may be repurposed from virtual buttons to an informational display. As shown in FIG. **8**, in one functional mode, edge display **52** may display virtual button icons such as icons **80**. In the example of FIG. **8**, icons **80** include a “+” and a “−” symbol surrounded by a circular border. In this example, icons **80** may indicate portions of display **14** that may be touched or tapped in order to raise (“+”) or lower (“−”) the volume of audio output from device **10**. Audio output may be associated with the voice of a

caller on a cellular phone, with music or other medial playback from device **10**. Audio output may be output from speakers such as speaker **19** (FIG. **1**) or from speakers associated with headphones or other remotely connected speakers attached (using wired or wireless connections) to device **10**. This is merely illustrative. Virtual button icons **80** may be any suitable icon associated with any operational mode of device **10**. As another example, icons **80** may be a camera icon indicating the location of a virtual camera shutter button when device **10** is operated as in a picture capturing mode.

(53) As indicated by arrows **86**, a user of device **10** may swipe edge display **52** (e.g., a single swipe in a single direction, multiple swipes in multiple directions, etc.) using a finger. Swiping edge display **52** may change the function of edge display **52**. Changing the function of edge display **52** using a swipe of edge display **52** is merely illustrative. The function of edge display **52** may be changed using any suitable touch input to display **14** (e.g., single tap, multiple taps, pinching, circular motions, etc.)

(54) As shown in FIG. **8**, swiping edge display **52** may cause virtual buttons **80** to be replaced by an informational display such as informational display **82**. In the example of FIG. **8**, informational display **82** is a “NOW PLAYING” text display associated with a media file being played back to a user of device **10**. This is merely illustrative. Informational display **82** may be any text, image or other graphical display. If desired, informational display **82** may be a flashing display, may appear temporarily and return to a virtual button display, may scroll across edge display **52** in direction **84**, direction **88** or in a direction perpendicular to directions **84** and **88**.

(55) If desired, when changing operating modes of device **10** or when changing the function of edge display **52**, virtual button icons **80** may be repositioned to another portion of edge display **52** or to another edge display **52** as shown in FIG. **9**. In the example of FIG. **9**, virtual volume button icons **80** may occupy a first region of edge display **52** such as region **90**. When a user changes the function of edge display **52** or changes the mode of operation of device **10** (e.g., by swiping edge display **52** as indicated by arrows **86**), region **90** may be repurposed as an informational display or as a virtual button with a different function (e.g., a virtual camera shutter button). When a user changes the function of edge display **52** or changes the mode of operation of device **10**, virtual button icons **80** may be moved to a different portion of edge display **52** such as portion **92**. Moving icons **80** to portion **92** may allow device **10** to use region **90** for another purpose (e.g., as a camera shutter or informational display) while still providing a user of device **10** with the ability to (for example) change audio output volume.

(56) FIG. **10** is a cross-sectional side view of a portion of device **10** in the vicinity of a virtual button **52** having a biasing member **94**. In the example of FIG. **10**, biasing member **94** is a dome-shaped biasing member that pushes transparent button member **62** outward in direction **96** when the user releases pressure from button member **62**. Other types of biasing members may used if desired, such as spring-based biasing members or other biasing structures that bias button members such as button member **62**. The use of a dome-shaped biasing structure is merely illustrative. If desired, transparent button member **62** may include a conductive material such as conductive material **98**. Conductive material **98** may form a portion touch-sensitive circuit that detects a touch of button member **62** by a user of device **10**. Positioning biasing member **94** between flexible display **14** and button member **62** is merely illustrative. Because display **14** is flexible, biasing member **94** may be placed in a position behind flexible display **14** such as position **100**. In configurations in which biasing member **94** is positioned behind flexible display **14**, button member **62** may deform flexible display **14** when pushed by a user of device **10**. Deforming flexible display **14** with button member **62** may compress biasing member **94**. When released, biasing member **94** may push flexible display **14** against transparent button member **62** pushing button member **62** outward in direction **96**.

(57) As shown in FIG. **11**, touch-sensitive flexible display regions **52** on edge **24** of device **10** may have an associated lens for magnifying or otherwise altering the path of light **64** emitted by display

layer **14**A of display **14**. For example, lens **102** may magnify text or other information displayed on sidewall (edge) display **52**). Edge displays **52** formed from active portions of touch-sensitive flexible display section **14**AB may be formed in openings such as openings **18** in a structural member such as structural member **68**. Structural member **68** may be formed from a portion of housing **12**, may be formed from a portion of cover layer **14**C or may be formed from another structural member of device **10**. A lens such as lens **102** may be mounted in opening **18**. Lens **102** may be formed from a transparent material such as plastic, glass, or other transparent material. Light such as light **64** generated by display pixels in display layer **14**A may pass through lens **102** so that edge display **52** may be visible through lens **102**. Providing edge display **52** with a lens such as lens may provide a brighter edge display, may cause a virtual button to appear larger than its physical size or may otherwise enhance the function of edge display **52**.

(58) If desired, lens **102** may be formed as a portion of a transparent button member such as transparent button member **62**, as shown in FIG. **12**. Lens **102** may be formed as an integral portion of button member **62** or may be a separate lens member that is mounted to button member **62**. Providing device **10** with edge displays **52** that form virtual buttons having transparent button members **62** with lenses **102** may provide a brighter edge display or may cause a virtual button to appear larger than its physical size while providing a tactile indicator to a user of device **10** of the location of virtual button **52**.

(59) FIG. **13** shows how touch-sensitive flexible display **14**AB may be coupled to control circuitry such as control circuitry **120**. Control circuitry **120** may include storage such as flash memory, hard disk drive memory, solid state storage devices, other nonvolatile memory, random-access memory and other volatile memory, etc. Control circuitry **120** may also include processing circuitry. The processing circuitry of control circuitry **120** may include digital signal processors, microcontrollers, application specific integrated circuits, microprocessors, power management unit (PMU) circuits, and processing circuitry that is part of other types of integrated circuits.

(60) Control circuitry **120** may be used to run software on device **10**, such as internet browsing applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. Control circuitry **120** may be used to configure and operate display pixels and touch-sensitive elements associated with touch-sensitive flexible display **14**AB. For example, control circuitry **120** may be used to illuminate or inactivate portions of display **14** to create active and inactive regions. As another example, control circuitry **120** may be used to change the operating mode of device **10** and/or the functional mode of edge displays **52** based on touch-input to touch-sensitive flexible display **14**AB or other user input. When a user touches virtual button **52**, control circuitry **120** can take suitable action. As examples, contact between a user's finger or other external object and virtual button **52** may direct device **10** to take actions such as displaying information for a user, making a volume adjustment to media that is being played to the user, controlling media playback, taking an action associated with a wireless communications session, or otherwise taking suitable action.

(61) One or more virtual buttons such as virtual button **52** may be used to form volume adjustment switches (e.g., sliding controls), ringer buttons, on/off buttons, sleep buttons, customized buttons (e.g., buttons that are specific to a particular program or operating system that is running on device **10** and that change in real time during use of device **10**), etc. If desired, virtual buttons may be labeled with particular colors, patterns, icons, text, or other information to assist a user in identifying the function of the button.

(62) Touch-sensitive flexible display **14**AB may include front surface display portions **112** and one or more sidewall surface (edge) display portions **114**. Sidewall surface display portions **114** may, if desired, be separated from front surface display portion **112** by regions **118**. Regions **118** may be blocked from view using masking structures such as portions of a device housing or a patterned opaque masking layer. Regions **118** may be regions of unilluminated (inactive) pixels. Control circuitry **120** may be used to configure touch-sensitive flexible display **14**AB to have regions **118**

with inactive pixels. As shown in FIG. **13**, regions **118** may, if desired, include control circuitry such as conductive traces **122**. Conductive traces **122** may be electrically coupled to control circuitry **120**. Conductive traces **122** may be control lines for display pixels in regions **112** and **114**. Display **14** may have additional control circuitry (e.g., control lines, drive lines, etc.) along a peripheral edge of display **14**. Providing display **14** with control circuitry in regions **118** may reduce the area required for control circuitry on the peripheral edge of display **14**. Providing display **14** with control circuitry in regions **118** may allow multiple displays (i.e., displays **112** and **114**) to be coupled to control circuitry **120** using a common interconnect (e.g., a common flex circuit that interconnects display **14** with control circuitry **120**. Connecting multiple displays to control circuitry **120** using a common interconnect may help reduce the space required for interconnects, thereby reducing, for example, the size, production cost and complexity of devices such as device **10**.

(63) As shown in FIG. **13**, sidewall surface display portions **114** may include illuminated touch-sensitive regions **52**. Control circuitry **120** may be used to configure touch-sensitive flexible display **14**AB to illuminate pixels in illuminated touch-sensitive regions **52** and to turn off (or inactivate) remaining pixels in sidewall surface display portions **114**. Illuminating pixels in illuminated touch-sensitive regions **52** and inactivating other pixels in sidewall surface display portions **114** may separate front surface display portion **112** from illuminated touch-sensitive regions **52**. Illuminated touch-sensitive regions **52** may be configured to remain stationary in sidewall surface display portions **114** or may be repositioned in sidewall surface display portions **114** during normal use of device **10** using control circuitry **120**. If desired, illuminated touch-sensitive regions **52** may occupy all or substantially all of sidewall surface display portions **114**. Virtual buttons **52** may be reconfigured during use of device **10**. For example, device **10** may use sidewall region **114** to display a first set of buttons when operated in one mode and may use region **114** to display a second (different) set of buttons when operated in another mode.

(64) FIG. **14** is a perspective view of device **10** showing how virtual buttons **52** may form a portion of a gaming controller when device **10** is operated in a landscape position. As shown in FIG. **14**, sidewall surface **24** may include one or more virtual buttons **52**. During operation of device **10** in a gaming mode, portions of display **14** that are visible on front surface **22** of device **10** may display additional virtual buttons such as additional virtual buttons **126**. Virtual buttons **52**, additional virtual buttons **126** and other components of device **10** (e.g., accelerometers) may be used in combination to deliver user input to device **10** for gaming software applications. Virtual buttons **52** may be operated by a touch, swipe, multiple touches or other touch inputs to virtual buttons **52** and **126**. Virtual buttons **52** may be used separately (i.e., one at a time) or in combination (e.g., simultaneously) to produce different inputs to device **10**. Virtual buttons **126** may be used separately (i.e., one at a time) or in combination (e.g., simultaneously) to produce different inputs to device **10**. Virtual buttons **52** and **126** may be used separately (e.g., one at a time, two at a time, etc.) or in combination (e.g., combinations of two virtual buttons, three virtual buttons four virtual buttons, or more than four virtual buttons simultaneously) to produce different inputs to device **10**.

(65) In order to prevent erroneous inputs to virtual buttons such as virtual buttons **52** on a sidewall of device **10**, device **10** may be provided with one or more sensors such as proximity sensor **124**. As an example, while device **10** is in a gaming mode of operation, device **10** may be temporarily held in palm of a user, in a pocket of a user's clothing, may be held in an orientation typically associated with using device **10** during a cellular telephone call, etc. Sensors such as proximity sensor **124** may be used to determine whether a touch input to virtual buttons **52** is an intended touch by a finger or, for example, an unintended touch by the palm of a hand. This is merely illustrative. Control circuitry **120** (see FIG. **13**) may be configured to use other device components such as light sensors, motion sensors (accelerometers), capacitance sensors, etc. to determine the orientation of device **10** and the intent of a touch input to virtual buttons **52**. Software running on control circuitry **120** may be configured to accept input from components such as proximity sensor

124 or other components to determine whether a touch input to virtual buttons **52** is intended or unintended. Touch inputs to virtual buttons **52** that are determined to be unintended may be ignored.

(66) FIG. **15** is a side view of a portion of an illuminated touch-sensitive region **52** on a sidewall portion **24** of display **14** showing how illuminated touch-sensitive region **52** may be used to form one or more selection buttons **127** for selecting a software application to be run on device **10** using control circuitry **120**. As shown in FIG. **15**, selection buttons **127** may include illuminated icons **128** associate with selected software applications such as text messaging, calendar, camera, calculator, media player, web browser, email client, cellular telephone, or other software applications. A selected software application may be activated using a touch input to a portion of illuminated touch-sensitive region **52** associated with a selected selection button **127**. As an example, selecting a camera application button may cause selection buttons **127** on illuminated touch-sensitive region **52** to be replaced by function buttons associated with the selected application.

(67) As shown in FIG. **16**, function buttons such as function buttons **130** associated with a selected application may include touch buttons having icons **132** displayed that indicate the function of function button **130** and slider buttons such as slider button **130** that have multiple associated icons **132** that indicate a function associated with multiple positions of slider button **130**. In the example of FIG. **16**, function buttons **130** include camera function buttons such as a button for turning on, off or setting to automatic setting for a camera flash. Function buttons **130** may include a button with a camera icon **132** indicating a camera shutter button. Function buttons **130** may include a slider switch with a first position **136** for selecting (for example) a snapshot mode and a second position **138** for selecting (for example) a video mode. This is merely illustrative. Slider buttons **130** may have a continuously changing function with the continuously changing position of the slider button (e.g., a continuous zooming control, focus control, light level control, exposure control, etc.).

(68) As shown in FIG. **17**, function buttons **130** may display a portion of a calendar. Calendar function buttons **130** may include visible indicators **134** of an appointment associated with a selected, displayed calendar day. Function buttons **130** associated with a calendar application may include arrows for causing edge display **52** to display adjacent calendar days. Illuminated touch-sensitive region **52** may allow scrolling of calendar function buttons vertically or horizontally along illuminated touch-sensitive region **52** in response to a swipe input by a user of device **10** in a vertical or horizontal direction respectively. Some software applications such as media player applications may have associated lists in addition to or instead of function buttons **130**.

(69) As shown in FIG. **18**, illuminated touch-sensitive region **52** may display a list associated with a media player software application for device **10**. In the example of FIG. **18**, illuminated touch-sensitive region **52** displays a scrollable list of song titles **140**. Swiping edge display **52** in a vertical direction (as indicated by arrows **144**) may cause edge display **52** to display song titles **140** above or below the currently displayed song title **140** in the list of song titles. Swiping edge display **52** in a horizontal direction (as indicated by arrows **146**) may cause edge display **52** to display a different list associated with the same software application such as artist lists, album lists, playlist lists, video lists, genre lists, webcast lists, audio book lists, etc.

(70) In the example of FIG. **19**, swiping edge display **52** in a horizontal direction (as indicated by arrows **146**) may cause edge display **52** to display a list of artists **142** associated with media files stored on circuitry **120** of device **10**. After swiping edge display **52** to cause edge display **52** to display a list of artists **142**, swiping edge display **52** in a vertical direction (as indicated by arrows **144**) may cause edge display **52** to display artist names **142** above or below the currently displayed artist name **142** as shown in FIG. **20**. The examples of FIGS. **15**, **16**, **17**, **18**, **19**, and **20** are merely illustrative. In general, illuminated touch-sensitive regions **52** on a sidewall surface **24** of device **10** may be virtual buttons such as caps lock, shift, control, delete, page up/down, number lock,

function-specific buttons, escape, enter, multiply, add, divide, subtract, memory storage, clear, all clear, percent, square root, other calculator buttons, text messaging, calendar, calculator, media player, web browser, email client, cellular telephone, or other software applications, menu, ringer on/off, ringer on/off/vibrate, lock/unlock, call, an end-call, or any other button or other visual information display.

(71) The foregoing is merely illustrative of the principles of this invention and various modifications can be made by those skilled in the art without departing from the scope and spirit of the invention.

Claims

1. An electronic device, comprising: a display having an array of pixels including first, second, and third groups of pixels, wherein the pixels comprise organic light-emitting diode pixels and wherein the display comprises inactive areas without pixels; and an opaque masking material that separates the first and second groups of pixels, wherein the third group of pixels is configured to display an image between first and second portions of the opaque masking material and wherein the opaque masking material comprises black ink.
2. The electronic device defined in claim 1 wherein the opaque masking material has a round opening.
3. The electronic device defined in claim 1 wherein the display comprises a touch sensor that detects touch input on the first, second, and third groups of pixels.
4. The electronic device defined in claim 3 wherein the image changes in response to the touch input.
5. The electronic device defined in claim 1 wherein the opaque masking material is located in the inactive areas of the display.
6. The electronic device defined in claim 1 wherein the image comprises a virtual button.
7. The electronic device defined in claim 1 further comprising circuitry that is hidden from view by the opaque masking material.
8. The electronic device defined in claim 1 wherein the display comprises a flexible display having a bent edge.
9. The electronic device defined in claim 1 wherein the electronic device is configured to launch a camera application in response to touch input on the third group of pixels.
10. An electronic device, comprising: a display having an array of pixels that forms an active area; and an opaque masking material that forms an inactive area within the active area, wherein the opaque masking material has first and second portions separated by a gap, wherein a group of the pixels are configured to display an image in the gap, and wherein the opaque masking material has an opening that is bordered on all sides by the opaque masking material.
11. The electronic device defined in claim 10 further comprising a touch sensor configured to detect touch input on the group of the pixels.
12. The electronic device defined in claim 11 wherein the electronic device is configured to launch a camera application in response to the touch input.
13. The electronic device defined in claim 10 wherein the image comprises a virtual button, wherein the pixels comprise organic light-emitting diode pixels, and wherein the opaque masking material comprises black ink.
14. The electronic device defined in claim 10 wherein the display comprises a flexible display having a bent edge.
15. An electronic device, comprising: a display having first, second, and third active areas; an opaque masking layer that separates the first and second active areas, wherein the opaque masking layer has a gap, wherein the display comprises pixels in the third active area that are configured to display an image in the gap, wherein the opaque masking layer has an opening, and wherein the

opaque masking layer extends around an entire perimeter of the opening.

16. The electronic device defined in claim 15 wherein the image comprises a visible indicator.

17. The electronic device defined in claim 15 wherein the third active area is touch-sensitive.

18. The electronic device defined in claim 15 wherein the image is repositioned in response to user input.

19. The electronic device defined in claim 18 wherein the user input comprises touch input.
